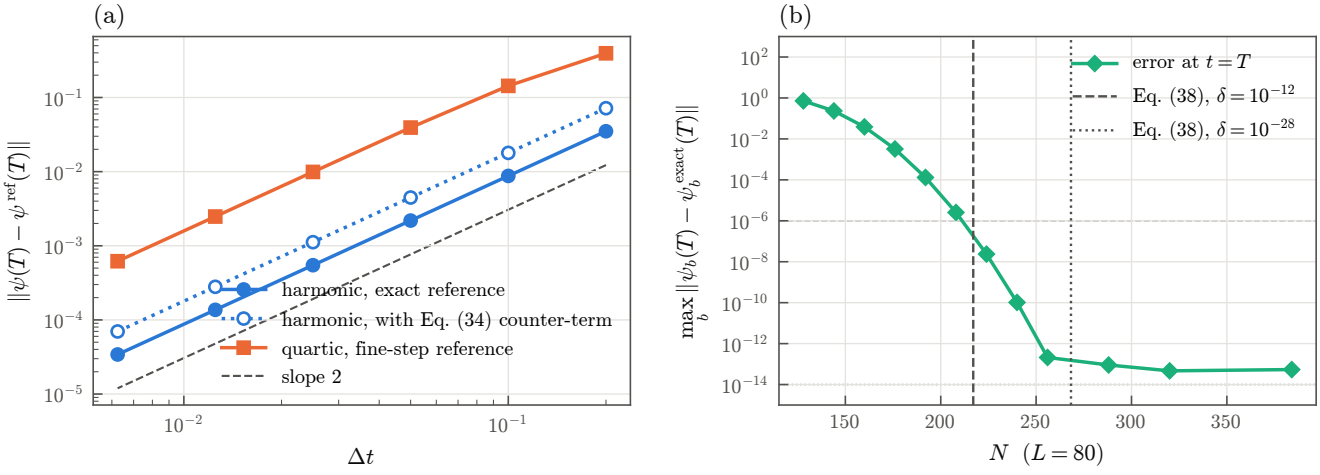
(a)
$\|\psi(T) - \psi^{\mathrm{ref}}(T)\|$
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-2}
10^{-1}
Δt
harmonic, exact reference
harmonic, with Eq. (34) counter-term
quartic, fine-step reference
slope 2
(b)
$\max_b \|\psi_b(T) - \psi_b^{\mathrm{exact}}(T)\|$
10^{2}
10^{0}
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
10^{-14}
150
200
250
300
350
N $(L = 80)$
error at $t = T$
Eq. (38), $\delta = 10^{-12}$
Eq. (38), $\delta = 10^{-28}$